PHY 614-2011
Problem Set #7

Problem 1 (7.4.2)

$$X = \frac{\sqrt{\hbar}}{\sqrt{2\omega m}}(a + a^\dagger) \qquad [a, a^\dagger] = 1$$

$$P = -\frac{i\omega\sqrt{\hbar}}{\sqrt{2\omega m}}(a - a^\dagger)$$

Since $\langle m|n\rangle = \delta_{mn}$, clearly

$$\langle X\rangle = \langle P\rangle = 0.$$

$$\langle X^2\rangle = \frac{\hbar}{2\omega m}\langle a^2 + a^{\dagger 2} + aa^\dagger + a^\dagger a\rangle$$

$$= \frac{\hbar}{2\omega m}\langle n|\, a^2 + a^{\dagger 2} + 2a^\dagger a + 1\,|n\rangle$$

$$\langle n|a^\dagger a|n\rangle = \sqrt{n}\,\langle n|a^\dagger|n-1\rangle$$
$$= n\,\langle n|n\rangle = n$$

$\Rightarrow$

$$\langle X^2\rangle = \frac{\hbar}{2\omega m}(2n+1) = \left(n+\tfrac{1}{2}\right)\frac{\hbar}{m\omega}$$

Sim $\langle P^2\rangle = m\hbar\omega\left(n+\tfrac{1}{2}\right)$

Note $\dfrac{\langle P^2\rangle}{2m} = \dfrac{1}{2}m\omega^2\langle X^2\rangle$

Thus $\langle(\Delta X)^2\rangle = \langle X^2\rangle = \left(n+\tfrac{1}{2}\right)\dfrac{\hbar}{m\omega}$

$\langle(\Delta P)^2\rangle = \langle P^2\rangle = \left(n+\tfrac{1}{2}\right)m\hbar\omega$

$$(\Delta X)(\Delta P) = \left(n+\tfrac{1}{2}\right)\hbar$$

Problem 3: (7.4.5)

(1) $|\psi(t)\rangle = \frac{1}{\sqrt{2}}\left(e^{-\frac{i\omega t}{2}}|0\rangle + e^{-\frac{3i\omega t}{2}}|1\rangle\right)$

(2) $\langle\psi(0)|x|\psi(0)\rangle$

$= \sqrt{\frac{\hbar}{2m\omega}}\,\frac{1}{2}\left(\langle 0| + \langle 1|\right)(a+a^{+})\left(|0\rangle + |1\rangle\right)$

$= \sqrt{\frac{\hbar}{2m\omega}}\cdot\frac{1}{2}\left(\langle 0|a|1\rangle + \langle 1|a^{+}|0\rangle\right)$

$= \sqrt{\frac{\hbar}{2m\omega}}$

Similarly $\langle\psi(0)|P|\psi(0)\rangle = \cancel{0\sqrt{\frac{m\omega\hbar}{2}}}\ 0$

$\langle\psi(t)|x|\psi(t)\rangle.$

$= \sqrt{\frac{\hbar}{2m\omega}}\,\frac{1}{2}\left(e^{\frac{i\omega t}{2}-\frac{3i\omega t}{2}} + c.c.\right).$

$= \sqrt{\frac{\hbar}{2m\omega}}\cos\omega t$

$\langle\psi(t)|P|\psi(t)\rangle$

$= i\left(\frac{m\omega\hbar}{2}\right)^{1/2}\frac{1}{2}\left(e^{\frac{3i\omega t}{2}-\frac{i\omega t}{2}} - c.c.\right)$

$= -\left(\frac{m\omega\hbar}{2}\right)^{1/2}\sin\omega t$

(3) Using Ehrenfest theorem

$\frac{d\langle x\rangle}{dt} = \frac{\langle P\rangle}{m} = -\frac{1}{m}\left(\frac{m\omega\hbar}{2}\right)^{1/2}\sin\omega t$

agrees with $\langle\psi(t)|x|\psi(t)\rangle$ above.

Problem 4:

(i) $a|\lambda\rangle = A_\lambda [a, e^{\lambda a^\dagger}]|0\rangle$
$+ A_\lambda e^{\lambda a^\dagger} a|0\rangle$

$= A_\lambda [a, e^{\lambda a^\dagger}]|0\rangle$

$$[a, e^{\lambda a^\dagger}] = \sum_{n=0}^{\infty} \frac{1}{n!} \lambda^n [a, a^{\dagger n}]$$

$$= \sum \frac{1}{n!} \lambda^n (n-1) (a^\dagger)^{n-1} = \lambda e^{\lambda a^\dagger}$$

$\Rightarrow$

$$a|\lambda\rangle = \lambda|\lambda\rangle$$

(ii) & (iii)

$$\langle\mu|\lambda\rangle = A_\lambda A_\mu^* \langle 0| e^{\mu^* \hat{a}} e^{\lambda \hat{a}^\dagger} |0\rangle$$

$$e^{\mu^* a} e^{\lambda a^\dagger} = e^{\lambda \hat{a}^\dagger} e^{\mu^* \hat{a}} e^{\mu^* \lambda [a, a^\dagger]}$$

$$= e^{\mu^* \lambda} e^{\lambda \hat{a}^\dagger} e^{\mu^* \hat{a}}$$

Use $e^{\mu^* a}|0\rangle = |0\rangle$ $\quad \langle 0| e^{\lambda a^\dagger} = \langle 0|$

$$\Rightarrow \langle\mu|\lambda\rangle = e^{\mu^* \lambda} A_\lambda A_\mu^*$$

$$\Rightarrow \langle\lambda|\lambda\rangle = e^{|\lambda|^2} |A_\lambda|^2$$

Thus $A_\lambda = e^{-\frac{1}{2}|\lambda|^2}$ up to a phase

$$\Rightarrow \langle\mu|\lambda\rangle = e^{-\frac{1}{2}(|\lambda|^2 + |\mu|^2 - 2\mu^* \lambda)}$$

(iv) Let $\hat{K} = \int \frac{d\lambda^* d\lambda}{2\pi} |\lambda\rangle\langle\lambda|$

Then

$$\langle\mu_1|\hat{K}|\mu_2\rangle = e^{-\frac{1}{2}(|\mu_1|^2+|\mu_2|^2)} \int \frac{d\lambda^* d\lambda}{2\pi} e^{-|\lambda|^2 + \mu_1^*\lambda + \lambda^*\mu_2}$$

$$\lambda \equiv \lambda_1 + i\lambda_2 \qquad \lambda_1, \lambda_2 \text{ real}$$

$$\Rightarrow \quad d\lambda^* d\lambda = 2\, d\lambda_1 d\lambda_2$$

Thus

$$\langle\mu_1|\hat{K}|\mu_2\rangle = e^{-\frac{1}{2}(|\mu_1|^2+|\mu_2|^2)} \int \frac{d\lambda_1 d\lambda_2}{\pi} e^{-\lambda_1^2 + \lambda_1(\mu_1^*+\mu_2)} e^{-\lambda_2^2 + i\lambda_2(\mu_1^*-\mu_2)}$$

$$= e^{-\frac{1}{2}(|\mu_1|^2+|\mu_2|^2)} \frac{1}{\pi}\sqrt{\pi}\, e^{-\frac{1}{4}(\mu_1^*+\mu_2)^2} \sqrt{\pi}\, e^{+\frac{1}{4}(\mu_1^*-\mu_2)^2}$$

$$= e^{-\frac{1}{2}(|\mu_1|^2+|\mu_2|^2 - 2\mu_1^*\mu_2)}$$

$$= \langle\mu_1|\mu_2\rangle$$

Since this is true for arbitrary μ_1, μ_2 we must have

$$\hat{K} = 1$$

(v) $\psi_\lambda(x)$ ~~satisfies~~ $= \langle x|\lambda\rangle$ satisfies

$$\frac{1}{\sqrt{2\hbar}}\left(x + \hbar\frac{\partial}{\partial x}\right)\psi_\lambda = \lambda\psi_\lambda$$

since $a = \frac{1}{\sqrt{2\hbar}}(x + ip)$

The solution is

$$\psi_\lambda(x) = N e^{-\frac{1}{2}\frac{x^2}{\hbar} + \sqrt{\frac{2}{\hbar}}\lambda x}$$

Use $\lambda = \frac{1}{\sqrt{2\hbar}}(x_0 + ip_0)$

$$\psi_\lambda = N e^{-\frac{1}{2}\frac{x^2}{\hbar} + \frac{1}{\hbar}(x_0 + ip_0)x}$$

To normalize

$$|N|^2 \int_{-\infty}^{\infty} dx\, e^{-x^2/\hbar}\, e^{\sqrt{\frac{2}{\hbar}}(\lambda+\lambda^*)x} = 1$$

$$\Rightarrow \quad |N|^2 \sqrt{\pi\hbar}\, e^{(\lambda+\lambda^*)^2/2} = 1$$

$$\Rightarrow \quad N = \frac{1}{(\pi\hbar)^{1/4}} e^{-(\lambda+\lambda^*)^2/4}$$ upto phase

Thus in terms of x_0, p_0

$$\psi_\lambda(x) = \frac{1}{(\pi\hbar)^{1/4}} e^{-\frac{1}{2\hbar}(x-x_0)^2 + \frac{i}{\hbar}p_0 x}$$

$\rightarrow$ Gaussian wave packet of width $\sqrt{\hbar}$ centered at x_0 and average momentum p_0

Problem 5:

$$|\lambda\rangle = e^{-\frac{1}{2}|\lambda|^2} e^{\lambda a^+} |0\rangle.$$

$$= e^{-\frac{1}{2}|\lambda|^2} \sum_{n=0}^{\infty} \frac{\lambda^n}{\sqrt{n!}} |n\rangle.$$

Since $(a^+)^n |0\rangle = \sqrt{n!}\, |n\rangle$

Thus

$$\psi_\lambda(x) = \langle x|\lambda\rangle = e^{-\frac{1}{2}|\lambda|^2} \sum_{n=0}^{\infty} \frac{\lambda^n}{\sqrt{n!}} u_n(x)$$

where $$u_n(x) = \left(\frac{\alpha}{\sqrt{\pi}\, 2^n n!}\right)^{1/2} e^{-\frac{1}{2}\alpha^2 x^2} H_n(\alpha x)$$

$$\alpha \equiv \sqrt{\frac{m\omega}{\hbar}}$$

Under time evolution

$$|n\rangle \to e^{-iE_n t/\hbar} |n\rangle$$

$$= e^{-i(n+\frac{1}{2})\omega t} |n\rangle$$

Thus

$$\psi_\lambda(x,t) = e^{-\frac{1}{2}|\lambda|^2} \sum_{n=0}^{\infty} \frac{\lambda^n}{\sqrt{n!}} e^{-i(n+\frac{1}{2})\omega t} u_n(x).$$

Use

$$H_n(z) = (-1)^n e^{z^2} \frac{d^n}{dz^n} e^{-z^2}$$

Define $y = \alpha x$

$$u_n(x) = \frac{1}{\sqrt{n!}} \left(\frac{\alpha}{\sqrt{\pi}\, 2^n}\right)^{1/2} (-1)^n e^{y^2/2} \frac{d^n}{dy^n} e^{-y^2}$$

Substituting

$\Rightarrow$

$$\psi_\lambda(x,t) = e^{-\frac{1}{2}|\lambda|^2} \sum_{n=0}^{\infty} \frac{(\lambda e^{-i\omega t})^n e^{-\frac{i}{2}\omega t}}{n!} (-1)^n \left(\frac{\alpha}{\sqrt{\pi}\, 2^n}\right)^{1/2} e^{y^2/2} \frac{d^n}{dy^n} e^{-y^2}$$

$$= \left(\frac{\alpha}{\sqrt{\pi}}\right)^{1/2} e^{\frac{1}{2}(y^2-|\lambda|^2)} e^{-i\omega t/2} \sum_{n=0}^{\infty} \frac{1}{n!}\left(-\frac{\lambda}{\sqrt{2}} e^{-i\omega t}\right)^n \frac{d^n}{dy^n} e^{-y^2}$$

For any smooth fn f

$$f(y-z) = \sum_{n=0}^{\infty} \frac{(-z)^n}{n!} \frac{d^n}{dy^n} f(y)$$

Thus

$$\psi_\lambda(x,t) = \left(\frac{\alpha}{\sqrt{\pi}}\right)^{1/2} e^{\frac{1}{2}(y^2-|\lambda|^2) - i\frac{\omega t}{2}} e^{-\left(y - \frac{\lambda}{\sqrt{2}} e^{-i\omega t}\right)^2}$$

The probability density is

$$\frac{\alpha}{\sqrt{\pi}} e^{y^2 - |\lambda|^2} e^{-\left[\left(y-\frac{\lambda}{\sqrt{2}}e^{-i\omega t}\right)^2 + \left(y - \frac{\lambda^*}{\sqrt{2}} e^{i\omega t}\right)^2\right]}$$

Writing $\lambda = \rho e^{i\delta}$. This becomes

$$\frac{\alpha}{\sqrt{\pi}} e^{-[y - \sqrt{2}\rho \cos(\omega t - \delta)]^2}$$

$\Rightarrow$ ~~Center~~

Thus the probability is a gaussian whose center is at

$$x_0 = \sqrt{2}\,\frac{\rho}{\alpha}\cos(\omega t - \gamma)$$

amplitude is $\sqrt{2}\,\frac{\rho}{\alpha} = \frac{\sqrt{2}}{\alpha}|\lambda|$

Phase is $\gamma = \arg(\lambda)$

Problem 6:

Since $V(x) = V(-x)$ we can treat even and odd parity eigenfunctions separately. We do odd parity – the even parity is similar

(i) For odd parity

$$\psi(x) = B \sin kx \qquad x < a$$

$$\psi(x) = Ce^{ikx} + De^{-ikx} \qquad x > a$$

Matching conditions at $x = a$

$\Rightarrow$

$$B \sin ka = Ce^{ika} + De^{-ika}$$

$$ik(Ce^{ika} - De^{-ika}) - kB \cos ka = \lambda B \sin ka$$

Thus

$$\frac{Ce^{ika} - De^{-ika}}{Ce^{ika} + De^{ika}} = \frac{\lambda \sin ka + k \cos ka}{ik \sin ka}$$

Defining $$F \equiv \frac{\lambda \sin ka + k \cos ka}{\sin ka}$$

$$C\left(1 - \frac{F}{ik}\right)e^{ika} = D\left(1 + \frac{F}{ik}\right)e^{-ika}$$

$$\Rightarrow \qquad |C| = |D|$$

Writing
$$C = \frac{A}{2i} e^{i\alpha}$$
$$D = -\frac{A}{2i} e^{-i\alpha}$$

We get for $x > a$
$$\psi(x) = A \sin(kx + \alpha)$$

(ii) The matching eqns at $x = a$ now become

$$B \sin ka = A \sin(ka + \alpha)$$
$$B(\lambda \sin ka + k \cos ka) = Ak \cos(ka + \alpha)$$
$$\Rightarrow \boxed{\frac{1}{k} \tan(ka + \alpha) = \frac{\sin ka}{\lambda \sin ka + k \cos ka}}$$

This determines α

(iii). The above eqns also imply
$$A^2 = B^2 \left(\sin^2 ka + \left(\frac{\lambda}{k} \sin ka + \cos ka\right)^2\right)$$
$$\Rightarrow \boxed{\frac{A^2}{B^2} = 1 + \frac{\lambda^2}{k^2} \sin^2 ka + \frac{\lambda}{k} \sin 2ka}$$

(iv) When $\lambda \gg k$

$$\frac{A}{B} \sim \frac{\lambda}{k}$$

except when $\sin ka$ is small.

This can happen if

$$ka = n\pi + \epsilon \qquad \epsilon \ll 1$$

Then $\qquad n = 1, 2, \ldots$

$$\frac{A^2}{B^2} \approx 1 + \frac{\lambda^2}{k^2}\epsilon^2 + \frac{2\lambda}{k}\epsilon$$

$$= \left(1 + \frac{\lambda\epsilon}{k}\right)^2$$

Thus A/B can vanish when

$$\epsilon = -k/\lambda$$

which is consistent with $\epsilon \ll 1$ and $\lambda/k \gg 1$

$$\Rightarrow \quad ka = n\pi - \frac{k}{\lambda}$$

$$\Rightarrow \quad \boxed{k = \frac{n\pi}{a + \frac{1}{\lambda}}}$$

For even parity solutions

$$+B' \cos ka = A' \cos(ka+\alpha')$$

$$-kA' \sin(ka+\alpha') + kB' \sin ka = \lambda B' \cos ka$$

Thus

$$A' \cos(ka+\alpha') = B' \cos ka$$

$$A' \sin(ka+\alpha') = B' \sin ka - \frac{\lambda B' \cos ka}{k}.$$

$\Rightarrow$

$$A'^2 = B'^2 \left[1 + \frac{\lambda^2}{k^2} \cos^2 ka - \frac{\lambda}{k} \sin 2ka \right]$$

Now A'/B' can be small only when

$$ka = (n+\tfrac{1}{2})\pi + \epsilon$$

Then

$$\cos ka = -(-1)^n \epsilon$$

$$\sin ka = (-1)^n$$

$$\frac{A'^2}{B'^2} = 1 + \frac{\lambda^2}{k^2}\epsilon^2 - \frac{2\lambda}{k}\epsilon = \left(1 - \frac{\lambda\epsilon}{k}\right)^2$$

$$\Rightarrow \quad \epsilon = \frac{k}{\lambda}$$

$$\Rightarrow \quad ka = (n+\tfrac{1}{2})\pi + \frac{k}{\lambda}$$

$$\Rightarrow \quad k\left(a - \frac{1}{\lambda}\right) = (n+\tfrac{1}{2})\pi$$